WORKED SOLUTION

# STEP I 2006, Question 3

Pure • Roots of Polynomials

**STEP I 2006, Question 3**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q3

## The key idea

Think about the graph, not the formula. A curve that starts below the axis and must end above it has to cross, and the value at $x = 0$ (the constant term) tells you which side of the axis the curve is on at the origin. For the cubic, the turning points do the work: the curve crosses the axis three times exactly when its local maximum is above the axis and its local minimum is below it.

## Solution

### Part (i)

Let $f(x) = x^2 + bx + c$. The graph is a parabola opening upwards, so $f(x) \to +\infty$ as $x \to \pm\infty$. If $c < 0$ then $f(0) = c < 0$: the curve is below the axis at $x = 0$.

Since $f$ is continuous, is positive for large negative $x$ and is negative at $0$, it crosses the axis somewhere in $(-\infty, 0)$. In the same way it crosses again in $(0, \infty)$. These two roots lie on opposite sides of $0$, so they are different. A quadratic has at most two roots, so the equation has exactly two distinct real roots, one negative and one positive. Hence $c < 0$ is a *sufficient* condition.

It is *not necessary*. For example, $x^2 - 3x + 2 = (x - 1)(x - 2)$ has the distinct real roots 1 and 2, but $c = 2 > 0$. (The condition that is both necessary and sufficient is $b^2 > 4c$; when $c < 0$ this holds automatically, because $b^2 \geq 0 > 4c$.)

### Part (ii)

We claim that the equation has two distinct positive roots if and only if

$$b^2 > 4c, \qquad b < 0, \qquad c > 0.$$

*Necessary.* Suppose the roots are $\alpha \neq \beta$ with $\alpha, \beta > 0$. Then $x^2 + bx + c = (x - \alpha)(x - \beta)$,

so

$$c = \alpha\beta > 0, \qquad b = -(\alpha+\beta) < 0, \qquad b^2 - 4c = (\alpha+\beta)^2 - 4\alpha\beta = (\alpha-\beta)^2 > 0.$$

*Sufficient.* If $b^2 > 4c$ the roots $\alpha, \beta$ are real and distinct. Their product is $c > 0$, so they have the same sign. Their sum is $-b > 0$, so that common sign is positive.

**Ask yourself** Why aren't $b < 0$ and $c > 0$ enough on their own?

Take $x^2 - x + 1$: here $b = -1 < 0$ and $c = 1 > 0$, but $b^2 - 4c = -3 < 0$, so there are no real roots at all. In graph terms, $c > 0$ puts the curve above the axis at $x = 0$ and $b < 0$ puts the vertex at $x = -\frac{b}{2} > 0$. We also need the vertex to lie below the axis: its height is $c - \frac{b^2}{4}$, which is negative exactly when $b^2 > 4c$. Each of the three conditions is needed.

### Part (iii)

Let $g(x) = x^3 + px + q$, so $g'(x) = 3x^2 + p$. A real cubic has three roots counted with multiplicity, and its non-real roots come in complex conjugate pairs. Note also that $g(0) = q < 0$ in both cases.

**The case $p > 0$, $q < 0$.** Here $g'(x) = 3x^2 + p > 0$ for every $x$, so $g$ is strictly increasing and has exactly one real root. Since $g(0) = q < 0$ and $g$ is increasing, the root is positive. The remaining two roots are a pair of non-real complex conjugates.

So there is one real root, which is positive and simple, and two non-real complex conjugate roots.

**The case $p < 0$, $q < 0$.** Write $p = -3a^2$ with $a > 0$, that is, $a = \sqrt{-p/3}$. Then $g'(x) = 3(x^2 - a^2)$, so $g$ is increasing on $(-\infty, -a]$, decreasing on $[-a, a]$ and increasing on $[a, \infty)$. There is a local maximum at $x = -a$ and a local minimum at $x = a$, with

$$g(-a) = -a^3 + 3a^3 + q = 2a^3 + q, \qquad g(a) = a^3 - 3a^3 + q = q - 2a^3.$$

Since $q < 0$, the local minimum $g(a)$ is negative. On $[0, a]$, $g$ decreases from $g(0) = q < 0$, so it has no roots there. On $[a, \infty)$ it increases from $g(a) < 0$ to $+\infty$, so it has exactly one root greater than $a$, which is positive. Any other real root must lie in $(-\infty, -a]$ or $(-a, 0)$, so it is negative, and whether such roots exist depends only on the sign of the local maximum $g(-a)$.

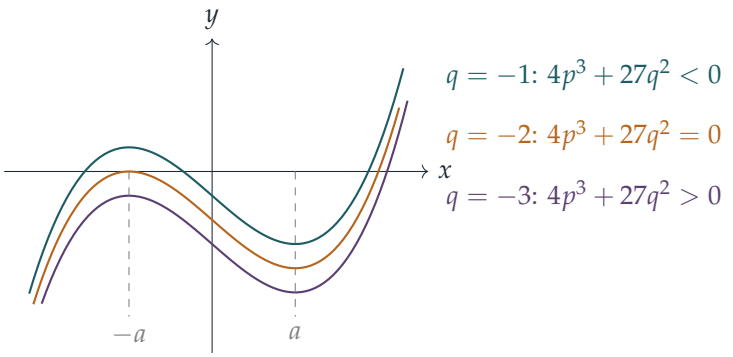
**Ask yourself** How does the sign of $g(-a)$ connect to $4p^3 + 27q^2$?

Multiply the two turning values:

$$g(-a)\,g(a) = (q + 2a^3)(q - 2a^3) = q^2 - 4a^6 = q^2 - 4\Big(-\frac{p}{3}\Big)^3 = \frac{4p^3 + 27q^2}{27}.$$

Since $g(a) < 0$, the sign of $g(-a)$ is the *opposite* of the sign of $4p^3 + 27q^2$. There are three cases.

- $4p^3 + 27q^2 > 0$. Then $g(-a) < 0$, so $g < 0$ on all of $(-\infty, a]$ (it increases up to the negative value $g(-a)$, then decreases). The only real root is the positive one. As before, there is one positive real root and two non-real complex conjugate roots.
- $4p^3 + 27q^2 = 0$. Then $g(-a) = 0$ and $g'(-a) = 0$, so $x = -a$ is a repeated root. Since the sum of the roots is 0 (there is no $x^2$ term), the third root is $2a$. The roots are $-a$, $-a$, $2a$: a negative double root and a positive simple root, all real.
- $4p^3 + 27q^2 < 0$. Then $g(-a) > 0$. Since $g(x) \to -\infty$ as $x \to -\infty$, there is a root in $(-\infty, -a)$. Since $g(-a) > 0 > g(0)$, there is another in $(-a, 0)$. Together with the root greater than $a$, this gives three distinct real roots: two negative and one positive.




The sketch shows $y = x^3 - 3x + q$ (so $p = -3$, $a = 1$) for $q = -1, -2, -3$: raising the local maximum through the axis turns one real root into three.

In every case with $q < 0$ there is exactly one positive root; this is consistent with the product of the roots being $-q > 0$.

## Checks

- With $p = -3$, $q = -1$ we get $4p^3 + 27q^2 = -108 + 27 = -81 < 0$, and numerically the roots are about $-1.532$, $-0.347$ and $1.879$: two negative, one positive, as predicted.
- With $p = -3$, $q = -2$: $4p^3 + 27q^2 = 0$ and $x^3 - 3x - 2 = (x+1)^2(x-2)$, giving roots $-1, -1, 2$.
- A numerical root-finder applied to thousands of random $(p, q)$ in each case, and a test of random $(b, c)$ against the conditions in (ii), agreed with every claim above.

## Common slips

- In (ii), giving only $b < 0$ and $c > 0$ (which ensure that *if* the roots are real they are positive) and forgetting $b^2 > 4c$.
- Proving only necessity, or only sufficiency, in (ii). "Necessary and sufficient" needs both directions.
- In (iii), forgetting the boundary case $4p^3 + 27q^2 = 0$, where there is a repeated root.

## Beyond STEP

The quantity $-(4p^3 + 27q^2)$ is the *discriminant* of the cubic $x^3 + px + q$, the exact analogue of $b^2 - 4ac$. It equals the product of the squared differences of the roots, so it vanishes exactly when there is a repeated root and is positive exactly when all three roots are real and distinct. Discriminants, and the way roots relate to coefficients, lead into Galois theory and the question of which equations can be solved by radicals. Allenby's *Rings, Fields and Groups* is a good place to start.

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